```python
# ========================================
# House Price Prediction using Linear Regression
# Google Colab Compatible
# ========================================

# Step 1: Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Step 2: Create Sample Dataset
data = {
    'Area_sqft': [1000, 1200, 1500, 1800, 2000, 2200, 2500, 2700, 3000, 3500],
    'Bedrooms': [2, 2, 3, 3, 3, 4, 4, 4, 5, 5],
    'Bathrooms': [1, 2, 2, 2, 3, 3, 3, 4, 4, 5],
    'Age': [15, 12, 10, 8, 7, 5, 4, 3, 2, 1],
    'Price': [3000000, 3500000, 4200000, 5000000, 5600000,
              6200000, 7000000, 7600000, 8500000, 9500000]
}

df = pd.DataFrame(data)

print("House Dataset")
print(df)

# Step 3: Define Features and Target
X = df[['Area_sqft', 'Bedrooms', 'Bathrooms', 'Age']]
y = df['Price']

# Step 4: Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Step 5: Train the Model
model = LinearRegression()
model.fit(X_train, y_train)

# Step 6: Predict Test Data
y_pred = model.predict(X_test)

# Step 7: Evaluate Model
print("\nMean Absolute Error:", mean_absolute_error(y_test, y_pred))
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Step 8: Predict Price for a New House
new_house = pd.DataFrame({
    'Area_sqft': [2100],
    'Bedrooms': [3],
    'Bathrooms': [3],
    'Age': [5]
})

predicted_price = model.predict(new_house)

print("\nPredicted House Price: ₹{:,.0f}".format(predicted_price[0]))

# Step 9: Display Model Coefficients
print("\nModel Coefficients:")
for feature, coef in zip(X.columns, model.coef_):
    print(f"{feature}: {coef:.2f}")

print("Intercept:", model.intercept_)

# Step 10: Visualization
plt.figure(figsize=(6,5))
plt.scatter(y_test, y_pred)
plt.plot([min(y_test), max(y_test)],
         [min(y_test), max(y_test)], 'r--')
plt.xlabel("Actual House Price")
plt.ylabel("Predicted House Price")
plt.title("Actual vs Predicted House Prices")
plt.grid(True)
plt.show()
```

```
House Dataset
   Area_sqft  Bedrooms  Bathrooms  Age    Price
0       1000         2          1   15  3000000
1       1200         2          2   12  3500000
2       1500         3          2   10  4200000
3       1800         3          2    8  5000000
4       2000         3          3    7  5600000
5       2200         4          3    5  6200000
6       2500         4          3    4  7000000
7       2700         4          4    3  7600000
8       3000         5          4    2  8500000
9       3500         5          5    1  9500000

Mean Absolute Error: 136280.8842652794
Mean Squared Error: 32931935822.61952
R2 Score: 0.9947308902683809

Predicted House Price: ₹5,944,148

Model Coefficients:
Area_sqft: 2534.46
Bedrooms: -36671.00
Bathrooms: 23602.08
Age: -22821.85
Intercept: 775097.5292587774
```